1. $u^*(0) = \frac{3}{2}, \quad u^*(1) = 1, \quad u^*(2) = \frac{1}{2}, \quad u^*(3) = 0,$

   $x^*(0) = 0, \quad x^*(1) = \frac{3}{2}, \quad x^*(2) = \frac{5}{2}, \quad x^*(3) = 3,$

   $\text{máx} J = J_0^*(x(0)) = \frac{15}{2}, \quad J_1^*(x(1)) = \frac{25}{4}, \quad J_2^*(x(2)) = \frac{29}{4}, \quad J_3^*(x(3)) = 4.$

2. $u^*(0) = \frac{3}{2}, \quad u^*(1) = 3,$

   $x^*(0) = 1, \quad x^*(1) = \frac{9}{2}, \quad x^*(2) = \frac{21}{2},$

   $\text{máx} J = J_0^*(x(0)) = \frac{47}{4}, \quad J_1^*(x(1)) = \frac{23}{2}, \quad J_2^*(x(2)) = \frac{21}{2}.$

3. $u^*(t) = \frac{1}{2}.\left(\frac{2}{3}\right)^t.\frac{1}{x_0}, \quad \text{para} \ \ t = 0, 1, 2, \ldots, T-1,$

   $x^*(t) = \left(\frac{3}{2}\right)^t, \quad \text{para} \ \ t = 0, 1, 2, \ldots, T,$

   $\text{máx} J = J_0^*(x(0)) = \frac{T}{3} + \ln\left[\left(\frac{3}{2}\right)^T.x_0\right], \quad J_t^*(x(t)) = \frac{(T-t)}{3} + \ln\left[\left(\frac{3}{2}\right)^T.x_0\right], \quad \text{para} \ \ t = 0, 1, 2, \ldots, T.$

4. $u^*(t) = \dfrac{1}{\displaystyle\sum_{k=0}^{T-t}\left[\frac{\rho}{(1+r)^2}\right]^k}, \quad \text{para} \ \ t = 0, 1, 2, \ldots, T-1,$

   $$x^*(t) = \left(\frac{\rho}{1+r}\right)^{2t}.\left\{\frac{\displaystyle\sum_{k=0}^{T-t}\left[\frac{\rho}{(1+r)^2}\right]^k}{\displaystyle\sum_{k=0}^{T}\left[\frac{\rho}{(1+r)^2}\right]^k}\right\}.x_0, \quad \text{para} \ \ t = 0, 1, 2, \ldots, T,$$

   $\text{máx} J = J_0^*(x(0)) = \sqrt{x_0}.\sqrt{\displaystyle\sum_{k=0}^{T}\left[\frac{\rho}{(1+r)^2}\right]^k}, \quad J_t^*(x(t)) = \left(\frac{1}{(1+r)}\right)^t.\sqrt{x(t)}.\sqrt{\displaystyle\sum_{k=0}^{T-t}\left[\frac{\rho}{(1+r)^2}\right]^k},$ para $t = 0, 1, 2, \ldots, T.$

5. $u^*(t) = \dfrac{1}{\displaystyle\sum_{k=0}^{T-t}\rho^k}, \quad \text{para} \ \ t = 0, 1, 2, \ldots, T-1,$

   $$x^*(t) = \rho^{2t}.\left\{\frac{\displaystyle\sum_{k=0}^{T-t}\rho^k}{\displaystyle\sum_{k=0}^{T}\rho^k}\right\}.x_0, \quad \text{para} \ \ t = 0, 1, 2, \ldots, T,$$

   $\text{máx} J = J_0^*(x(0)) = \sqrt{x_0}.\sqrt{\displaystyle\sum_{k=0}^{T}\rho^k}, \quad J_t^*(x(t)) = \sqrt{x(t)}.\sqrt{\displaystyle\sum_{k=0}^{T-t}\rho^k}, \quad \text{para} \ \ t = 0, 1, 2, \ldots, T.$

6. $x^*(t) = \frac{1}{3}.e^{-3t}$ realiza el mínimo del funcional.

7. $x^*(t) = \dfrac{4e^4-1}{e^8-1}.e^{2t} + \dfrac{e^8-4e^4}{e^8-1}.e^{-2t}$ realiza el mínimo del funcional.

8. $x^*(t) = \dfrac{6,6e^6-6}{e^{12}-1}.e^{3t} + \dfrac{6e^{12}-6,6e^6}{e^{12}-1}.e^{-3t}$ realiza el máximo del funcional.

9. $x^*(t) = \dfrac{1}{4}t^3 + \dfrac{1}{2}t + 1$ realiza el mínimo del funcional.

10. $x^*(t) = \dfrac{2e-1}{e^2-1}.e^{t} + \dfrac{e^2-2e}{e^2-1}.e^{-t}$ realiza el máximo del funcional.

11. $x^*(t) = \dfrac{x_0.e^{\frac{2}{c}T}}{e^{\frac{2}{c}T}-1}.e^{-\frac{1}{c}t} - \dfrac{x_0}{e^{\frac{2}{c}T}-1}.e^{\frac{1}{c}t}$ realiza el mínimo del funcional.

    Si $\quad c \to 0^+ \quad$ la extremal es $x^*(t) = \begin{cases} x_0, & \text{si } t = 0 \\ 0, & \text{si } t > 0. \end{cases}$

12. $x^*(t) = \dfrac{1}{12}t^3 - \dfrac{12(x_0-x_1)+T^3}{2(T-1)}t + x_0 + \dfrac{12(x_0-x_1)+T^3}{2(T-1)}$

    $u^*(t) = \dfrac{1}{4}t^2 - \dfrac{12(x_0-x_1)+T^3}{2(T-1)}$

    $\lambda^*(t) = -\dfrac{1}{2}t^2 + \dfrac{12(x_0-x_1)+T^3}{(T-1)}$

13. $x^*(t) = e^t.\cos(\sqrt{2}t) + x_1.e^{-\frac{5\pi}{2\sqrt{2}}}e^t.\mathrm{sen}(\sqrt{2}t)$

    $u^*(t) = (1+\sqrt{2}.x_1.e^{-\frac{5\pi}{2\sqrt{2}}}).e^t.\cos(\sqrt{2}t) + (x_1.e^{-\frac{5\pi}{2\sqrt{2}}} - \sqrt{2}).e^t.\mathrm{sen}(\sqrt{2}t) = \lambda^*(t)$

14. $x^*(t) = \dfrac{1}{4}t^2 + \dfrac{5}{2}t$

    $u^*(t) = -\dfrac{1}{2}t - \dfrac{5}{2}$

    $\lambda^*(t) = -t - 5$